Peyman Ghaediarjenaki

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Home: Tehran (Iran)

Education

Master of Science in Energy System Engineering Nov 2021 – Feb 2024

- Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran
- Thesis title**:** Intelligence fault detection and classification in photovoltaic systems using machine learning techniques
- GPA**:** 16.42 out of 20, GPA (Including thesis): 17.1 out of 20
- Relevant coursework: *Advanced Mathematical Programming/Process Engineering/Modeling of Energy Systems/Energy Economics*

Bachelor of Science in Electrical Engineering Sep 2016 – July 202

- Arak University, Arak, Iran
- Thesis title**:** Simulation of soft switched non-isolated high step-down (DC-DC) converter with PSpice software
- GPA**:** 14.80 out of 20, GPA (The last 60 hours): 16.38 out of 20
- Relevant coursework: *Electrical Energy Generation/Electric Energy Systems Analysis1,2/Operaation Research*

Research Interests

- Energy Yield Assessment
- Photovoltaic Systems
- Energy Transitions
- PV System Modelling
- Hydrodynamic Modelling
- Techno-economic Optimization

Academic Publications

Journal: **Ghaedi P, Eskandari A, Nedaei A, Habibi M, Parvin P, Aghaei M. Ensemble LVQ Model for Photovoltaic Line-to-Line Fault Diagnosis Using K-Means Clustering and AdaGrad. Energies. 2024; 17(21):5269**. doi.org/10.3390/en17215269. (Citation: 6)

- *Proposed an LVQ ensemble that detects PV LL faults under critical high-impedance/low-mismatch conditions with **99.26%** accuracy.*
- *Led I–V-based feature engineering and experiments and wrote the paper as **first author**.*

Journal: **Ghaedi P, Eskandari A, Nedaei A, Hatami F, Parvin P, Aghaei M. Logically Optimized and Probabilistic Integrated Photovoltaic Fault Finding Package based on Machine Learning. Frontiers in Energy Research. 2025; Volume 13**. doi: 10.3389/fenrg.2025.1675953.

- *Built a **fuzzy-logic + PSO ensemble** that integrates ML classifiers for multiple fault types.*
- *Designed the **multi-fault dataset** and benchmarking, demonstrating superior accuracy over recent PV fault-diagnosis methods.*

Journal: **Mamershafai R, Ghaedi P, Moradi Sizkouhi A, Talebi S, Aghaei M. Bird Dropping Fault Detection in Photovoltaic Panels via Stable Diffusion-Generated Data and Optimized Deep Learning Models** (under review – to Energy journal).

- *Co-developed the stable-diffusion-based **image generation** and augmentation pipeline for bird-dropping faults in **PV panel images**.*
- *Contributed to training and hyperparameter tuning of **CNN** and **YOLO** models and to interpreting **≈95%** bird-fault detection accuracy.*

Journal: **Ghaedi P, Eskandari A, Nedaei A, Aghaei M. Sequential Fault Detection and Resistance Estimation in PV Arrays Using Deep Q-Network and Bayesian Neural Networks** (Prepared to submit).

- *Developed a two-stage PV line-to-line fault pipeline that detects faults and estimates fault-path resistance (0–100 Ω) with **99.2% R^2** on testing.*
- *As first author, led I–V-curve feature design, dataset generation, and training of the **RL**-based classifier and **Bayesian resistance estimator**.*

Conference: **Nedaei A, Eskandari A, Salehpour S, Ghaedi P, Shafie-khah M , Aghaei M. AI in Smart Grids for Enhanced Renewable Energy Management: Part 1. Techniques and Applications** (accepted – FES Conference 2025).

Conference: **Nedaei A, Eskandari A, Salehpour S, Ghaedi P, Shafie-khah M, Aghaei M. AI in Smart Grids for Enhanced Renewable Energy Management: Part 2. Challenges and Future Directions** (accepted – FES Conference 2025).

- *Co-authored and **presented** Part 1 (Techniques and Applications) at FES 2025, Cluj-Napoca (online).*
- *Co-authored and **presented** Part 2 (Challenges and Future Directions) at FES 2025, Cluj-Napoca (online).*

Technical & Research Experience

Optimization of PV Tilt Angle for Shahrekord Using PVsyst Data and ODE-Based Numerical Methods in Python *[link]***.**

Independent research project September 2025

- *Configured a fixed-tilt PV system in PVsyst for Shahrekord and **exported 8,760 hourly irradiance** and climate samples into **Python**.*
- *Developed an **ODE-based** Python model to compute solar geometry, plane-of-array irradiance, and PV power for arbitrary **tilt angles**.*
- ***Optimized** annual **energy yield** as a scalar function of tilt angle, identifying a fixed tilt that delivers a several-percent gain over rule-of-thumb settings.*
- *Validated Python energy estimates against PVsyst simulations and **plotted energy-versus-tilt** curves to support reproducible PV design decisions.*

CFD Analysis of Hydrodynamic Forces on a Circular Cylinder in Water ***[link]***.

Independent research project — September 2025

- *Simulated external flow around a circular cylinder in water using **2D CFD** to evaluate **hydrodynamic** loading across relevant Reynolds numbers.*
- *Generated **boundary-layer** and wake-refined meshes with inlet/outlet and no-slip wall conditions for stable incompressible solutions.*
- *Integrated surface pressure and shear to compute drag and **lift** forces and the **drag** coefficient.*
- *Benchmarked **drag and wake patterns** against canonical cylinder data, validating the CFD model for design use.*

Simulation of DC Electrical Faults in PV Systems Using MATLAB for Data Generation ***[link]***.

Independent research project — September 2024

- *Built a MATLAB/Simulink model of the PV DC side (array, DC–DC converter, DC load) to simulate **normal operating** conditions.*
- ***Parameterized** PV and converter models with irradiance, temperature, and fault settings to generate realistic DC voltage and current waveforms.*
- *Injected open-circuit, line-to-line, and gradual degradation faults along PV strings and the DC link, capturing transient and steady-state responses.*
- *Logged and labeled key signals into structured **normal/fault datasets** for training and benchmarking **data-driven PV** fault detection algorithms.*

Deep Neural Network–Based Parameter Estimation of PV One-Diode Models from I–V Curve Key Points ***[link]***.

Independent research project — September 2024

- *Reformulated one-diode PV parameter extraction as a **regression** problem using compact I–V key-point and **slope features** instead of full I–V curves.*
- *Generated synthetic datasets by sampling realistic parameter sets and simulating I–V curves under different irradiance and temperature conditions.*
- *Trained and tuned a fully connected **DNN** in Python (**TensorFlow**) to robustly map feature vectors to model parameters.*
- *Evaluated performance via I–V curve reconstruction and RMSE, enabling fast, non-iterative parameter extraction for PV monitoring and diagnosis.*

Simulation and Optimization of National-scale Energy Systems in EnergyPLAN ***[link]***.

Course project for modeling of energy systems — May 2023

- *Modeled a national-scale reference energy system in EnergyPLAN and built policy/technology scenarios to assess primary energy use and CO_2.*
- *Applied a **50% reduction** in residential space-heating demand and analyzed impacts on peak heat load, primary energy consumption, and emissions.*
- *Replaced conventional district-heating boilers with CHP, waste-heat recovery, and thermal storage to evaluate gains in overall fuel efficiency.*
- *Integrated large-scale **offshore wind**, a central heat pump, and thermal storage and compared CHP dispatch strategies to study system flexibility.*

Application of the Simplex Method to Reduce the Peak Demand and Cost of Household Consumers ***[link]***.

Course project for energy economics — November 2022

- *Formulated a **linear-programming** model for household appliance scheduling to **minimize** peak demand and electricity cost via the Simplex method.*
- *Implemented the optimization in MATLAB using realistic residential loads, tariffs, and appliance time windows.*
- *Achieved noticeable reductions in simulated **household peak power** and daily **electricity bills** compared to the unscheduled baseline.*
- *Demonstrated how optimized household scheduling can support **demand-side management** and peak shaving at scale.*

Entropy Analysis and Exergy Calculation of all Flows in a Power Plant Cycle ***[link]***.

Course project for optimization of exergy flow — December 2021

- *Modeled each component of a steam power-plant cycle (**turbines, pumps, heat exchangers, compressors**) in **MATLAB** and **Python** using thermodynamic properties from **CoolProp**.*
- *Calculated entropy generation, exergy destruction, and second-law efficiency for all control volumes to build a complete exergy balance of the cycle.*
- *Cross-validated MATLAB and Python/CoolProp results to check numerical consistency and robustness of the **thermodynamic** calculations.*
- *Ranked components by **exergy destruction** to identify major irreversibilities and highlight opportunities for improving overall **cycle efficiency**.*

Optimum Design of a Photovoltaic Power Plant Connected to the Grid in Tehran Using PVsyst Software ***[link]***.

Course project for power generation — November 2020

- *Designed a 100 kW **grid-connected PV** plant in PVsyst for Tehran, selecting modules and inverters compatible with local climate and grid codes.*
- *Sized strings and array layout within inverter DC/AC limits to **maximize** annual energy yield and ensure reliable operation.*
- *Analyzed shading patterns, array losses, performance ratio, and **annual energy production** in PVsyst to compare alternative configurations.*
- *Identified a cost- and land-efficient final design and documented key indicators (**specific yield, PR, loss diagram**) as a template PV project.*

Work Experience

Technical Expert in Power Engineering — Sep 2025 – Nov 2025

Noavaran Rah Saadat, Tehran, Iran ***[link]***

- *Prepared and reviewed comprehensive engineering documentation for solar PV power plant projects.*
- *Developed and evaluated technical drawings and plans using **AutoCAD** with strong accuracy and attention to detail.*
- *Applied advanced knowledge of LVAC, MVAC, and DC standards to perform precise **electrical calculations** and ensure compliance within **solar power plant systems**.*

R&D Engineer Feb 2024 – Sep 2024

Solar Tabesh Tavan BNL Company, Tehran, Iran ***[link]***

- *Designed and implemented **machine learning** models for automated **fault detection** and performance degradation analysis in PV modules and inverters.*
- *Analyzed **operational data** (voltage, current, power, temperature) to detect anomalous patterns and **predict failures** proactively.*
- *Conducted research on advanced techniques and methodologies for **monitoring** and **optimizing PV** system performance.*

Research Assistant Sep 2021 – Feb 2024

Master Thesis, Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran ***[link]***

- *Reviewed existing literature and state-of-the-art methods for DC electrical fault detection in photovoltaic systems, covering **signal processing**, power-loss analysis, and **AI-based approaches**.*
- *Proposed robust and efficient methodologies and identified critical gaps in **ensemble learning** models applied to photovoltaic fault detection.*
- *Compared and evaluated the **proposed** techniques against previously published methods to validate performance improvements.*

Engineering Intern May 2021 – Aug 2021

Raad Industrial Group, Shahrekord, Iran ***[link]***

- *Gained **hands-on experience** in the design and assembly of **electronic boards** (PCBs).*

Honors & Awards

Admitted to Amirkabir University of Technology through the Nationwide University Entrance Exam with a tuition-waiver scholarship, ranking in the top 1% among 25,000 candidates in the national master's entrance exam in electrical engineering.

- *The competition is intense since it is the only means to gain admission to universities.*

Admitted to Arak University through the Nationwide University Entrance Exam with a tuition-waiver scholarship, ranking in the top 3% among 180,000 candidates in Iran's national undergraduate entrance exam.

- *The competition is intense since it is the only means to gain admission to universities.*

Skills

English

Duolingo English Test Test score: 110 Test date: 2025-01-09

Programming

Python (libraries: NumPy, Pandas, Matplotlib, Scikit-Learn, TensorFlow, PyTorch), MATLAB, HTML, CSS

Software

PVsyst, DIgSILENT, Altium designer, Arduino, PSpice, Microsoft Office (Word, Excel, PowerPoint and Visio), LaTeX

AI & Data Science

Machine Learning & Deep Learning**:** Classification (MLP, SVM, KNN, LR, GNB, and DT), Regression (SVR, DNN, XGboost, Adaboost), Clustering (K-Means), Ensemble Learning, Reinforcement learning (DQN), Bayesian Neural Networks

Computer Vision**:** Image Classification (CNN: VGG, ResNet, DenseNet), Object Detection (YOLOv8, YOLOv9, YOLOv10), GAN, Stable Diffusion Models

Modeling and Optimization

GA, PSO, Fuzzy Logic, Convex Optimization, MILP, Linear Programming

References

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